DATA SCIENCE

If I Have 0 Knowledge of Machine Learning, This Is My 60-Day Plan

A day-by-day 60-day plan from zero to building and explaining real machine learning models. 1.5 to 2 hours a day. Free resources only.

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How the 60 days are split

Each block builds on the one before. Do not skip ahead.

- **Days 1-10: Python and Pandas**
  You cannot do ML without handling data.
- **Days 11-20: Statistics and maths you need**
  Only the parts ML actually uses.
- **Days 21-35: Core algorithms**
  Regression, classification, trees, clustering.
- **Days 36-45: Making models good**
  Features, validation, tuning, imbalanced data.
- **Days 46-60: Two projects**
  The part that gets you interviews.

Daily rule: 30 minutes learning, the rest coding. Watching without coding does not count.

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Days 1 to 10: Python and Pandas

Goal: load any CSV, clean it, and summarise it without help.

- **Days 1-3**

  Python basics: variables, lists, dicts, loops, functions.

- **Days 4-6**

  Pandas: read_csv, info, filtering with loc and query, groupby.

- **Days 7-8**

  Cleaning: missing values, duplicates, types, dates.

- **Days 9-10**

  Charts with Matplotlib or Seaborn. Mini task: explore one Kaggle dataset and write 5 findings.

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Days 11 to 20: Statistics and maths

Learn what each idea means and when it matters.
Skip proofs.

- **Days 11-13**

  Mean, median, standard deviation, distributions, percentiles.

- **Days 14-15**

  Correlation, and why it is not causation.

- **Days 16-17**

  Probability basics and conditional probability.

- **Days 18-19**

  Linear algebra intuition: vectors, matrices, dot product. Just enough to read model code.

- **Day 20**

  Gradient descent intuition: how a model improves by small steps. A visual explainer is enough.

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Days 21 to 35: Core algorithms

Learn each with scikit-learn on a real dataset.

```
from sklearn.model_selection import (
    train_test_split)
X_tr, X_te, y_tr, y_te = train_test_split(
    X, y, test_size=0.2)
model.fit(X_tr, y_tr)
print(model.score(X_te, y_te))
```

Days 21-23

Train-test split, linear regression, MAE and RMSE. Predict house prices.

Days 24-26

Logistic regression, confusion matrix, precision, recall, F1. Predict churn.

Days 27-29

Decision trees and random forests. Feature importance.

Days 30-32

Gradient boosting with LightGBM or XGBoost.

Days 33-35

K-Means clustering. Segment customers and name each segment.

Days 36 to 45: Making models good

This is where beginners and job-ready candidates separate.

- **Days 36-37**

  Cross-validation. Why one train-test split is not enough.

- **Days 38-39**

  Feature engineering: encoding categories, scaling, date features.

- **Days 40-41**

  Imbalanced data: class weights, resampling, choosing the right metric.

- **Days 42-43**

  Hyperparameter tuning with GridSearchCV or RandomizedSearchCV.

- **Days 44-45**

  Pipelines: put cleaning and model in one object so nothing leaks.

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Days 46 to 60: Two projects

One classification, one regression or clustering. Real business framing.

- **Days 46-52: Project 1, churn prediction**

  Clean, explore, baseline, improve, explain top features, estimate ₹ impact of acting on predictions.

- **Days 53-58: Project 2, sales forecast or segmentation**

  Same structure. Finish with clear recommendations.

- **Day 59**

  Push both to GitHub with a README a manager can read in 2 minutes.

- **Day 60**

  Practise explaining each project out loud in 2 minutes.

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Free resources for each block

Use one per block. Switching between many slows you down.

- **Python and Pandas**

  Kaggle Learn: Python and Pandas courses.

- **Statistics**

  StatQuest (YouTube): short visual explanations.

- **Algorithms**

  scikit-learn user guide plus StatQuest videos on each algorithm.

- **Practice data**

  Kaggle datasets: telecom churn, house prices, mall customers.

Day 60 check: can you take a new dataset, build a baseline and an improved model, and explain why it is better? Then you are ready to apply.

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